Becoming: χ Before the Clock, and the Zero-Divisor Conjecture

Harald Kjøge Rønning

A companion to Foundation, Revision 21. Built entirely from its proven results; new content tagged by its own honest weight.

Revision 13.1. Three clauses plus one micro-fix, all confirmed against the actual text before editing. §2's new [SCOPE] note misattributed the completion endpoint's timelessness to "the symmetric limit" — wrong mechanism (the actual argument is the freeze dichotomy, §2's own first [FACT]) and wrong vocabulary ("symmetric" already names the *other* endpoint elsewhere in this corpus, Foundation's own "symmetric reference" for the unity end). Corrected to name the freeze dichotomy directly. §4's new [FACT, cited] block said the coupling $B = [L_u, R_w] \circ \tau$ "is exactly the associator" — the companion document's own abstract states $[L_u, R_w](y) = -(u, y, w)$, minus the associator, not the associator itself; spectrally irrelevant since β sees only singular values, but a flat identity claim should match its source's sign. Corrected. The same block's norm-bound gloss, "attained with equality on that locus," blurred a set of elements with the pairs the equality condition actually concerns; corrected to name pairs. The [LEDGER]'s flag that the grading-ansatz item "closed in Foundation Revision 25.1" is imprecise — the closure was recorded in Revision 25; 25.1 repaired a sentence describing it. Corrected to "closed in Foundation Revision 25, current 25.1."

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1. χ_{kin}: the domain before the clock

[PROP, extending an existing definition, not a new result] Foundation defines χ via the symplectic spectrum of a state's covariance operator, applied directly, and separately via constancy across an entire folium — every state normal to a family member's world gets that member's value, by definition of what a sector label is. On states with covariance close to, but not exactly, a family member's — a small Hilbert–Schmidt-class perturbation, still quasi-equivalent, still in the same folium — the two conventions disagree: folium-constancy assigns the family value; direct evaluation assigns whatever the perturbed covariance's own spectrum gives, generically different. This is a real contradiction, not a stylistic tension, and it is resolved by naming two different objects rather than picking one.

[DEF] χ_{sector} is Foundation's original quantity: folium-constant, a macrostate label — distinct values mark disjoint representations, full stop, regardless of which state within a folium is being asked about.

[DEF] χ_{kin} is a different, finer-grained quantity: the symplectic spectrum evaluated directly on whatever covariance a state actually has, varying *within* a folium as the state fluctuates away from its sector's equilibrium representative. The two relate simply: χ_{sector} is χ_{kin} evaluated at the folium's own equilibrium point; χ_{kin} fluctuates around that value for other, non-equilibrium members of the same folium. Macrostate depth and microstate depth — not a patch on the contradiction, but arguably the more physical picture underneath it.

[FACT] On the equilibrium spine, one parameter β coordinates every mode's symplectic eigenvalue coherently, so χ_{kin} reduces there to a single number, $\nu(\beta)$, recovering χ_{sector} exactly. Off the spine, a generic quasi-free covariance has no reason to be coordinated by one shared parameter at all — its symplectic spectrum is a function of mode index, not a scalar — so two generic states are, in general, *incomparable* in depth: χ_{kin} is spectrum-valued off the spine, and "depth" there is a partial order, not a single dial. This was not previously noticed and is worth carrying forward explicitly rather than glossed as "a measure of depth" without qualification. It is also not a defect peculiar to this framework: ordinary thermodynamics has the same structure, temperature being a single well-defined number only at equilibrium, and something coarser (energy distributed however it happens to be distributed, not summarized by one parameter) off it. A partial order off the spine is thermodynamic realism, not a weakness of the coordinate.

A measure of depth that exists before clock-time is exactly this: χ_{kin} asks nothing about temperature to be evaluated on any single covariance, because depth was never actually a thermal notion — β was only ever how the equilibrium spine happens to coordinate itself. Its extension to χ_{sector}, the family-level label, is the object with the folium semantics; χ_{kin} itself is defined pointwise, needing no folium, no equilibrium, and — off the spine — no single number to report.

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2. Two timeless boundaries, one interior

[FACT, cited] The tracial stratum ($\beta \to 0$) is flowless: by the freeze dichotomy, flow trivial $\iff$ tracial, and no admissible configuration is ever tracial.

[FACT, cited] Every genuine member of the family, at any finite β, is by definition a KMS state for the derived flow at that β — hence carries a nontrivial, well-defined modular flow, hence, on the Connes–Rovelli thermal-time reading, a genuine internal clock.

[PROP] The pure, unsqueezed vacuum is not a member of the family at any finite β; it is the $\beta \to \infty$ limit of $\nu(\beta) \to \frac{1}{2}$. KMS states of a nontrivial dynamics are never pure at finite β — purity is exclusively the zero-temperature limit — so the pure vacuum is excluded from carrying a clock for the same reason it is excluded from the family at all: it is not a finite-β member, only an unattained limit of one.

[PROP, restated as alignment, not existence] Clock-time exists far more generally than the family: the Connes–Rovelli thermal-time hypothesis assigns a modular clock to *every* faithful normal state, family member or not — this is the content of Tomita–Takesaki theory applied honestly. What is special about the family's interior is not that a clock exists there and nowhere else, but that every interior member's own private modular clock *aligns*, up to a rescaling by β, with the one derived dynamics. Off the family, on states with no equilibrium relationship to that dynamics, a state still carries its own modular clock — just a private one, not synchronized to anything external. "Settling," on this reading, is better understood as the alignment of private clocks into a shared flow, with the family as the fully-aligned locus, than as the appearance of time where none existed.

[FACT] The unity end's timelessness is established via the corner construction — already Foundation's own convention for non-faithful states elsewhere: restrict to the pure state's own support projection; the corner it generates is one-dimensional; modular theory on a one-dimensional algebra is trivially, unproblematically well-defined, and trivially gives the identity flow. This is the route that works. Applying Tomita–Takesaki directly to the pure state's full GNS representation does not: that construction needs the GNS vector to be both cyclic and separating for the algebra it generates, and a pure state's GNS vector is cyclic but never separating in dimension greater than one.

[INTERPRETATION] Read this way, unsharp unity acquires an unplanned, structurally identical twin: unsharp completion. Neither end was ever a configuration the universe occupied; both are boundary limits of an interior that is itself uniformly clocked throughout. "Window" was always the wrong word for something with no edges to be a window's edges — the right image is an open interior whose closure adds two points it never contains, not a discrete threshold.

[SCOPE, global versus local] Both timeless endpoints established above — unity via the corner construction, completion via the freeze dichotomy at the tracial limit — are statements about the *global* operational algebra specifically, the one algebra this construction builds from all of $\mathrm{Im}(\mathrm{Fix}(\tau))$ at once. Nothing here extends to any local subalgebra a future construction might carve out of it — a bounded region's own algebra, in the sense this project's later interior-net work builds. That extension does not merely need checking; the standard expectation runs the other way: local algebras of bounded regions are generically type III_1 with genuinely nontrivial modular flow (Bisognano–Wichmann for wedges, Hislop–Longo for diamonds) regardless of how trivial the global state's own endpoint behavior is — precisely the structure this project's own interior-net paper independently establishes for its reconstructed net, citing the same two results. A local observer restricted to such a region would not, on general grounds, inherit either global endpoint's timelessness. This is a scope boundary on the argument above, not a live tension with it: the corner construction says something true and specific about the global algebra, and says nothing at all, one way or the other, about any local algebra not yet constructed here.

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3. The conservation theorem, reread

[FACT, cited] Because χ is built from symplectic eigenvalues, it is exactly conserved by any symplectically implemented dynamics — the derived squeeze flow included, for every state it transports.

[INTERPRETATION] Read against P-Em: the ground's activity lies outside the vocabulary of algebraic representation; the conservation theorem is this claim's appearance as mathematics. Any dynamics the formalism can host as an automorphism group is, by the theorem, incapable of changing depth — so depth-changing becoming cannot appear in the formalism as an automorphism. Not an absence to apologize for. The axiom's own signature, showing up as a proof.

[FACT] Nothing in the conservation theorem, by itself, forbids the derived flow from moving a state across distinct, χ-equal sectors — the theorem forbids only changing χ's value, which is a narrower claim than forbidding all motion between sectors. The precise statement is **per-χ-fiber, not per-sector**. Separately, and independently of this theorem: the flow moves nothing between family members at all, χ-equal or not, because every family member is individually stationary under it (below confirms this directly, and confirms there are no χ-equal, squeeze-distinct sectors within the family for the theorem's narrower reading to apply to in the first place). What the conservation theorem alone establishes is depth-invariance; what the family's own structure separately establishes is that there is nothing else to move between.

[DEF] χ_{depth} names the coordinate already formalized — $\nu(\beta)$, conditional on the thermodynamic-limit [TARGET] in Foundation. χ_{shape} consists of the mediator and split components — a coordinate on the orbit data Foundation's own [OPEN] question gestures at, the "which representative" that $\mathrm{Aut}(\mathbb{S})$-orbit language leaves free. The [FACT] below explains why squeeze is not a third component.

[FACT, cited precisely] The squeeze-thermal family has no parameter independent of β. The infrared write-out paper (§6, Scope) states directly that β "remains the one genuine physical parameter of the state"; the uniqueness paper (§2, "Uniqueness within the full quasi-free class") proves this at full strength — the KMS condition forces any anomalous pair-correlation content (the technical shape squeezing takes) to zero within the *entire* quasi-free class, not a restricted subclass. There is accordingly no generator of "motion between squeeze frames at fixed temperature" for any bracket computation to act on, because no such frames exist — squeeze was never a live degree of freedom here.

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4. The zero-divisor conjecture

[FACT, cited] Everything invertible in the current architecture descends from the composition stratum: the derived flow, $J = L_{e_4}$, the symplectic structure itself are all built from $\mathrm{Im}(\mathbb{O})$, where the norm still multiplies.

[FACT, cited to the published literature, corrected against primary sources] The doubling from $\mathbb{O}$ to $\mathbb{S}$ adds exactly two things beyond the composition stratum: the τ/μ spine, and the zero-divisor structure — elements whose left-multiplication maps are singular, non-invertible. This structure is independently established in the published literature, precisely stated: Moreno (1998, *Bol. Soc. Mat. Mexicana* (3) 4, 13–28) proves that the space of normalized *pairs* (x, y), both norm one, with $xy = 0$ — Moreno's own definition of "zero divisor," a pair rather than a single element — is isometric to the compact form of the exceptional Lie group G_2 with a naturally reductive left-invariant metric. The zero-divisor *locus* in the more familiar sense — individual elements admitting some nonzero annihilating partner — is a related but distinct object: more recent work (Reggiani, 2024, arXiv:2411.18881, building on Biss–Dugger–Isaksen) shows this locus is isometric to the Stiefel manifold $V_2(\mathbb{R}^7)$, the space of orthonormal 2-frames in $\mathbb{R}^7$, with a specific G_2-invariant metric — itself a homogeneous space under the G_2 action, its dimension (11) matching G_2's own (14) minus the 3-dimensional stabilizer of a fixed pair. de Marrais (2000, arXiv:math.GM/0011260) gives the explicit combinatorial structure — 42 "assessors," organized into box-kites — matching this project's own count. (The secondary literature's wording varies — de Marrais's own abstract says "homomorphic," not the correct term — so the statement above is pinned to the primary source and to which of the two objects, pairs or locus, it actually concerns.)

[CONJECTURE, speculative, theory-native, unverified] If depth-changing dynamics must be non-invertible — which §3's reading of the conservation theorem suggests without proving, since no such dynamics has been constructed — the zero-divisor structure is the only native candidate available: the sole place in $\mathbb{S}$'s own architecture where non-invertibility already exists, unforced, prior to any further construction. Stated at the weight it has earned and no further: **the generator of any depth-changing dynamics descends from $\mathbb{S}$'s zero-divisor structure, as the derived flow descends from $\mathrm{Im}(\mathbb{O})$'s composition structure.** The corrected citation above sharpens the resonance rather than weakening it: the pairs manifold governing the zero-divisor structure is not merely G_2-organized but *literally* the group whose remnant, via the same G_2, already governs the derived flow's own $\mathfrak{so}(3)$ symmetry elsewhere in this project — one group, doing both jobs. One caveat carried honestly rather than smoothed over: the passage this conjecture bets on — algebraic structure supplying a genuine dynamical generator — has exactly one proven instance anywhere in this framework, the derived flow's own descent from the composition stratum through a contraction (the standalone compilation, *The Keystone*, works this out in full), and no second. One precedent is real corroboration; it is not yet a pattern.

[FACT, cited] A companion document (*The Redistribution Operator*) has since identified the zero-divisor structure's own generating mechanism exactly, turning the pure geometry above into an explicit algebraic operator. For $A = v_1 + v_2 e_8$, the coupling governing $L_A^\top L_A$'s spectrum is $B = [L_u, R_w] \circ \tau$, for $u, w \in \mathrm{Im}(\mathbb{O})$ the imaginary parts of v_1, v_2 — the commutator equal to minus the associator map $y \mapsto (u, y, w)$, up to the sign convention recorded there, composed with conjugation. The spectrum splits into three bands — $N(A) - \beta$ (multiplicity 4), $N(A)$ (multiplicity 8, the associative middle band, where the associator vanishes identically by Artin's theorem), $N(A) + \beta$ (multiplicity 4) — with β an explicit closed form in u, w. Zero divisors are exactly, provably, the locus where this coupling *saturates*: $\beta = N(A)$ precisely there, the sharp bound $\|xy\| \leq \sqrt{2}\,\|x\|\|y\|$ attained with equality exactly at the pairs the zero-divisor geometry supplies, and nowhere else — proven, not observed.

[CAUTION] Any specific proposed shape for such a dynamics — a completely-positive semigroup, a dissipative generator — is imported scaffolding, standard in open-quantum-system theory, not implied by the algebra. The zero-divisor structure is offered here as a candidate *source*, not a candidate *form*; whether it produces something semigroup-shaped, or something else entirely, is exactly what remains to be checked, not assumed. A further distinction, sharper than either of these: naming a source is not yet naming a *mechanism* — a proven analogy carries an algebra-to-dynamics transfer step (contraction, rigorously understood), and this conjecture does not yet have one; it names where a generator might live, not how algebraic non-invertibility becomes dynamical irreversibility. A candidate mechanism *class* exists in the same native family the one proven instance belongs to, and the companion document above sharpens it from a located region into a computable direction: a singular scaling limit along the saturation locus specifically, where the lower spectral band $N(A) - \beta$ collapses to zero — not merely "near" the zero-divisor structure, but exactly the direction the associator's own saturation identifies, with the associative middle band ($N(A)$, coupling-free by Artin) remaining regular throughout any such limit. This sharpens *which directions degenerate*; it does not supply the transfer step itself — nothing in the companion document constructs a dynamical generator, semigroup or otherwise, and its own scope section says so explicitly, leaving the physical question — whether fermionic-style construction needs exact nilpotency or only the right statistics — exactly as open as it was. Offered here as the class worth trying first, at conjecture weight, not as an established transfer.

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5. The interpretive bridge, named

[NEW TAG: INTERPRETATION] Distinct from [PROP], [FACT], [POST], [CONJECTURE], and [TARGET]: this tag marks a claim connecting the formalism to a narrative or physical reading, where the connection itself — not any formal content on either side — is what remains unverified. Its introduction is a methodological addition to the project's tagging discipline, meant to carry forward into every future document, not only this one.

[INTERPRETATION] The family's β-axis is read as settling history — depth changing along cosmic time. This identification requires three things, none of which currently exist: a mechanism tracing an actual path through the family (§3's named absence); a proof that path is monotone in χ_{depth}; and a reason the universe's actual states are members, or local relatives, of *this specific* family rather than some other. It is currently carried by exactly one postulate (Foundation's [POST]) and two ledger items — not by any theorem.

[INTERPRETATION, with a stated caution — reframed as a deferred obligation, not a dissolved one] β, the modular parameter, is not radiation temperature, and the two should not be read as one axis under different names needing careful translation now. Ordinary radiation temperature presupposes a settled mass or energy scale to be measured against — precisely the kind of quantity this project's own earlier work shows is not yet meaningful vocabulary absent settled mass. β presupposes nothing external. So the apparent tension — the observed early universe was hot, χ's early end is cold in β — is not currently a paradox, because only one of the two axes is well-posed in the regime where the comparison would need to be made. But declaring it ill-posed now is not the same as declaring it resolved: the moment a mass sector exists, the Penrose-style reconciliation this document's [POST] already gestures at (hot in radiation, low in settling, because the relevant gravitational degrees of freedom were unsettled) becomes a *mandatory consistency check*, not an optional one. The honest tag for this item is a deferred obligation the future mass sector must meet, carried on the ledger below, not a pseudo-problem that has already been dissolved.

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6. What this document is

A statement of what follows from Foundation's proven results when read for their consequences, plus what those results suggest but do not show. Every claim above carries one of five weights: [FACT] (cited, already proven elsewhere), [PROP] (a short, direct consequence proven here), [DEF] (a definition, extending existing formalism), [CONJECTURE] (speculative, theory-native, named as such), or [INTERPRETATION] (a bridge between formalism and narrative, itself unverified). This document asserts nothing Foundation's current text denies.

[LEDGER] Open, in order of priority: the mechanism of settling itself, confirmed absent by two independent arguments — the conservation theorem, and every family member's own KMS-stationarity under the derived flow. χ_{shape}'s sole remaining question, now two small finite-dimensional computations rather than one (per Foundation's sharpened [OPEN]): the dimension of the S_3-invariant intersection of the three τ-anchored test spaces, since these are genuinely different subspaces of $\mathrm{Im}(\mathbb{S})$, not one shared algebra; and, on whatever core that intersection yields, whether μ commutes with the derived flow there or intertwines three copies of it — with a structural reason to expect intertwining over commutation. The zero-divisor conjecture, now two genuinely separate stages rather than one: (i) identify a mechanism class carrying algebraic non-invertibility into dynamical irreversibility — the conjecture as it stands names a source, sharpened by a companion document (§4) into a computable direction, the singular limit along the saturation locus specifically, but still not a transfer, and remains incomplete as a mechanism until one is proposed; (ii) only once a mechanism class is settled does computing what it actually produces become a well-posed question at all (§4). A further, unresolved fork this conjecture inherits rather than settles: whether algebraic descent from structure to dynamics is a construction device this framework uses to build things (epistemic — a modeling choice, with "why this shape" a classification question) or a claim that settling itself genuinely is contraction-like, ontic — in which case shape-selection and mechanism become one fused question rather than two. Neither branch is asserted; the fork is recorded so a future draft does not silently pick one. The three-part settling-history identification in full (§5). The mass-sector consistency check on β-versus-radiation-temperature, named as a deferred obligation rather than a dissolved question (§5). This document now draws directly on *The Keystone*'s compiled chain for the one proven algebra-to-dynamics precedent cited above, and on *The Redistribution Operator* for the zero-divisor mechanism's own exact algebraic identification (§4); its own header continues to pin Foundation, Revision 21, under this project's consumption policy — a sibling header tracks the parent revision whose content it actually cites, not the parent's latest revision number, and nothing Foundation has changed since Revision 21 is content this document depends on. This document inherits every open item of Foundation, Revision 21, without exception, and adds these. One inherited item has since closed, flagged here explicitly since the consumption policy above does not track it automatically: Foundation's grading-ansatz item — why the flow's underlying three-coordinate/three-momentum shape was assumed rather than derived — closed in Foundation Revision 25, current 25.1, via the same companion artifact now cited in Keystone (*The Non-Associativity Selector*), proving the shape forced rather than assumed. This document's own header pin is unchanged by that closure; the item is noted as resolved on its own terms, not as grounds to bump the pin.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.